

QUEUE

Q2)

Question:

DS	Session	Queue	Question Information	Level 1 Challenge 52
Question description There is a bit string consisting of n bits. Then, there are some changes that invert one given bit. Your task is to report, after each change, the length of the longest substring whose each bit is the same.				
Constraints • $1 \leq n \leq 2 \cdot 10^5$ • $1 \leq m \leq 2 \cdot 10^5$ • $1 \leq x_i \leq n$				
Input Format The first input line has a bit string consisting of n bits. The bits are numbered $1, 2, \dots, n$. The next line contains an integer m: the number of changes. The last line contains m integers $x_1, x_2, \dots, x_m$ describing the changes.				
Output Format After each change, print the length of the longest substring whose each bit is the same.				

PROGRAM:

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char q[200005];
6      int n, m, x;
7
8      scanf("%s", q);
9      n = strlen(q);
10
11     scanf("%d", &m);
12
13     while (m--) {
14         scanf("%d", &x);
15         x--;
16
17         if (q[x] == '0')
18             q[x] = '1';
19         else
20             q[x] = '0';
21
22         int max = 1, count = 1;
23
24         for (int i = 1; i < n; i++) {
25             if (q[i] == q[i - 1])
26                 count++;
27             else
28                 count = 1;
29
30             if (count > max)
31                 max = count;
32         }
33
34         printf("%d ", max);
35     }
36
37     return 0;
38 }
```

OUTPUT:

TEST CASE 1:

```
001011
3
3 2 5
4 2 3
```

TEST CASE 2:

```
101010
4
1 3 2 6
2 4 2 2
```